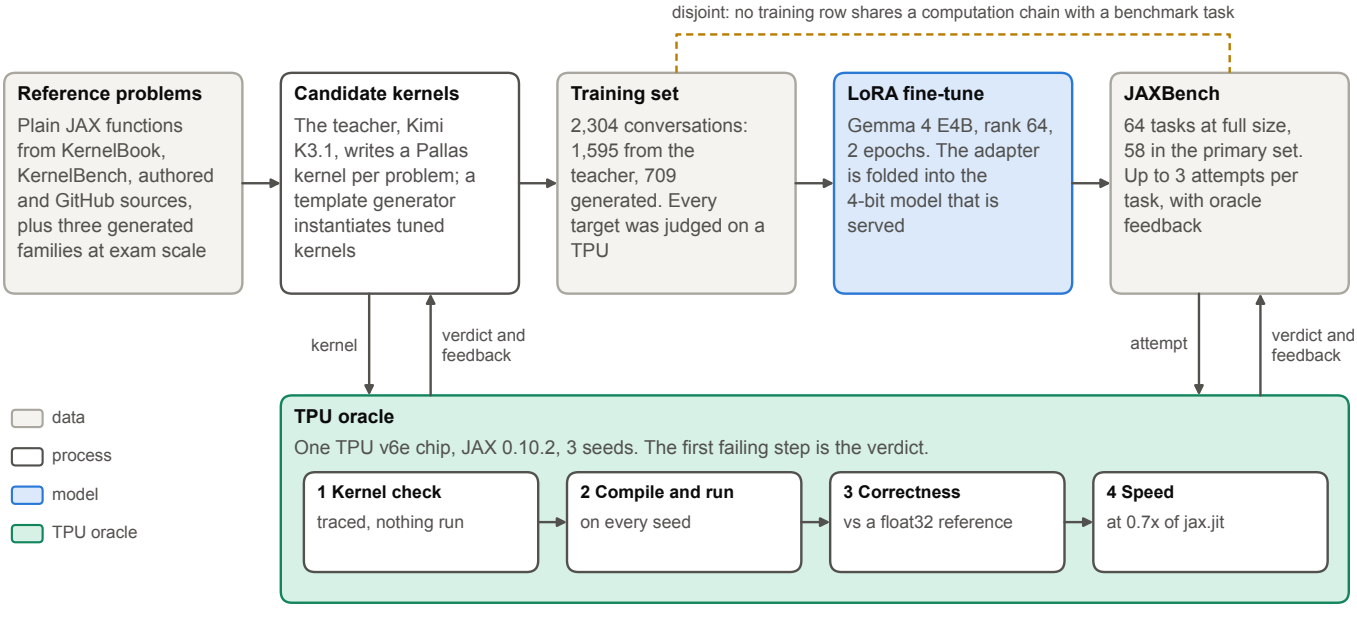
disjoint: no training row shares a computation chain with a benchmark task
Reference problems
Plain JAX functions from KernelBook, KernelBench, authored and GitHub sources, plus three generated families at exam scale
Candidate kernels
The teacher, Kimi K3.1, writes a Pallas kernel per problem; a template generator instantiates tuned kernels
Training set
2,304 conversations: 1,595 from the teacher, 709 generated. Every target was judged on a TPU
LoRA fine-tune
Gemma 4 E4B, rank 64, 2 epochs. The adapter is folded into the 4-bit model that is served
JAXBench
64 tasks at full size, 58 in the primary set. Up to 3 attempts per task, with oracle feedback
kernel
verdict and feedback
attempt
verdict and feedback
data
process
model
TPU oracle
TPU oracle
One TPU v6e chip, JAX 0.10.2, 3 seeds. The first failing step is the verdict.
1 Kernel check
traced, nothing run
2 Compile and run
on every seed
3 Correctness
vs a float32 reference
4 Speed
at 0.7x of jax.jit